

Network Traffic Analysis Report

Date: April 16, 2026

Tool Used: Wireshark

1. Executive Summary

This report provides a comprehensive analysis of network traffic captured using Wireshark. The objective was to examine communication patterns, identify protocols, and detect potential vulnerabilities such as unencrypted transmissions or suspicious connections. The findings indicate mostly normal activity with strong reliance on HTTPS, though minor risks exist.

2. Methodology

Traffic was captured and analyzed from a .pcapng file using Wireshark. Filters (HTTP, DNS) were applied. Statistical tools such as Conversations and Protocol Hierarchy were used. Packet inspection was performed to identify plaintext data and anomalies.

3. Findings

3.1 Protocols Observed

TCP, DNS, HTTP, and HTTPS were identified. HTTPS dominated traffic, indicating secure communication.

3.2 DNS Activity

DNS queries reflected normal browsing and background services. However, DNS traffic remains visible and unencrypted.

3.3 Traffic Analysis

Most traffic occurred between local and external IPs over ports 80 and 443. No abnormal spikes were detected.

3.4 Plaintext Data

No sensitive plaintext data such as credentials was found. Limited HTTP usage reduces exposure risk.

3.5 Suspicious Activity

No confirmed malicious activity was found. Some unknown IPs were observed but without abnormal behavior.

4. Risks & Recommendations

Risks

- HTTP exposure • DNS visibility • Unknown IP connections

Recommendations

- Enforce HTTPS • Enable DNS-over-HTTPS • Monitor outbound traffic • Use IDS/IPS tools

5. Conclusion

Traffic appears normal with no compromise detected. Security can be improved with better encryption and monitoring practices.